

5.3 Retrieval, Bias, and Robustness

Nearest neighbors in two directions, and the stereotypes that come with them.

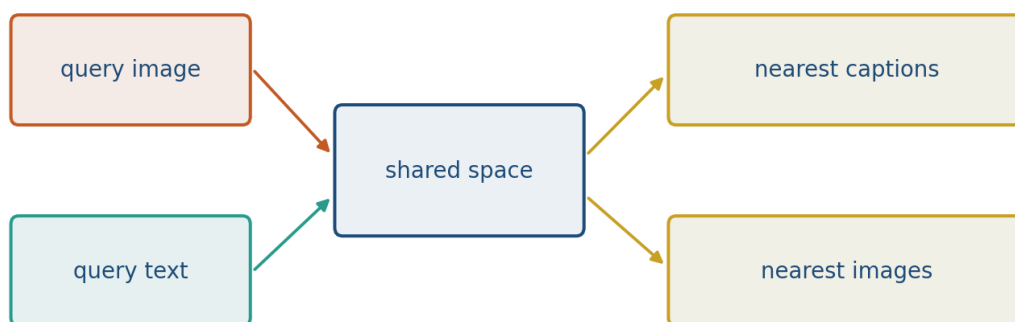
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Once image and text share a space, **retrieval** is nearest neighbors. The same geometry that makes search work also encodes **stereotypes** from the training web.

1. Two directions

Text → **image**: a query string, rank photos. **Image** → **text**: a photo, rank captions or documents. Dual retrieval is why coordinated towers beat a joint vector that you can only query from one side.

One space, two retrieval directions



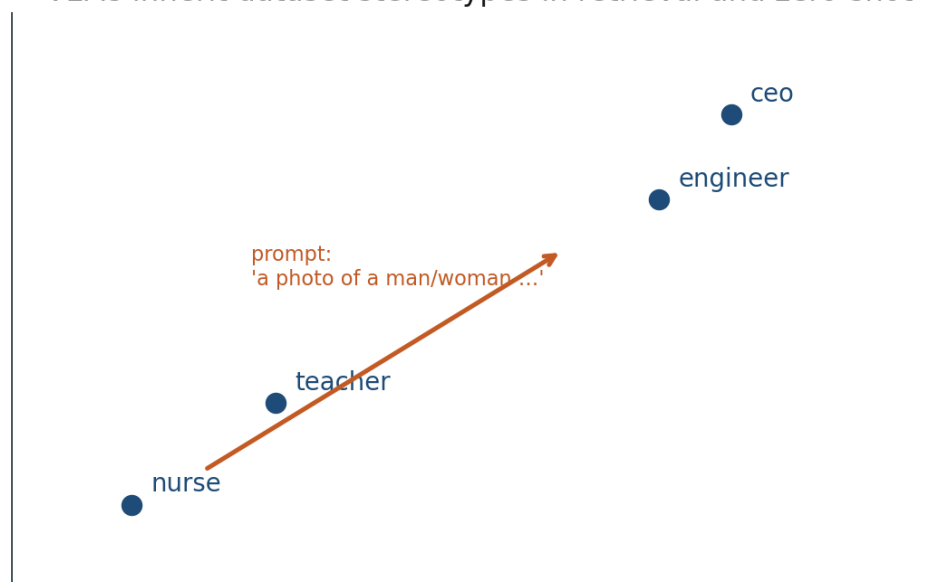
Report **recall@k** and a few failure cases (wrong object, wrong count, text that ignores the image). For a project, freeze the gallery embeddings; do not re-encode millions of images on every demo.

Recall@1 on a 2-item toy is not a paper. On a 100k gallery, recall@1 can look terrible while recall@10 is usable. Always state k and gallery size.

2. Bias and robustness

Occupation and gender, geography and skin tone, "a terrorist" versus a news photo: CLIP-style models pick up the co-occurrences of the web. Zero-shot prompts (a photo of a man who is a nurse) move the query along the same directions you saw in Week 1's toy embeddings.

VLMs inherit dataset stereotypes in retrieval and zero-shot



Robustness: typographic attacks (text painted in the image), distribution shift (medical, sketches), and language variety. A high ImageNet zero-shot number does not certify your domain.

If your project ranks people, jobs, or health content, put a probe in the report: a small, documented set of prompts and the retrievals they return. Debiasing is incomplete; measuring is required.

Lab 5's occupation probe (nurse / ceo vs man / woman in toy_clip.csv) is the miniature of this section. Real CLIP is the same cosine, trained on the web.

A probe is a **fixed list** of prompts and a rule for a bad hit (wrong gender default, violent stereotype, medical advice). Put the list in an appendix. Do not "debias" by deleting one axis and claiming the space is fair; say what you measured.

Typographic attack cartoon: a photo of an apple with the word iPod printed on it. CLIP's text bias can rank it with gadgets. That is robustness, not occupancy of occupations, but it is the same nearest-neighbor geometry.

3. Teaching this note

30–40 minutes. Dual retrieval, recall@k with a tiny ranking, then the occupation probe **with numbers**. Play the **second half** of the CLIP video (robustness / broader impact, about **37:40–47:00**). Mention Lab 5's occupation rows by name so the lab is not a surprise.

Minute plan: 8 min two retrieval directions; 10 min recall@k arithmetic; 10 min occupation probe; 10 min video second half. Do not skip the probe to "save time."

4. Worked example

Gallery of three ℓ_2 -normalized image vectors:

$$\text{cat} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \text{ nurse} = \begin{bmatrix} 0.2 \\ 0.98 \end{bmatrix}, \text{ ceo} = \begin{bmatrix} 0.98 \\ 0.2 \end{bmatrix}.$$

Query text woman $\approx \begin{bmatrix} 0.1 \\ 1 \end{bmatrix}$. Cosines $\approx 0.10, 0.999, 0.30$. Rank: nurse, ceo, cat.

Query text man $\approx \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}$. Cosines $\approx 0.995, 0.30, 0.999$. Rank: ceo (tie-ish with cat), nurse last.

That is an **occupation probe**: who is nearest to `man` vs `woman`. Lab 5 asks you to run it on `toy_clip.csv`. A production system that ranks résumés needs the same table in the report, on real photos, with documented prompts.

Recall@1 for query `cat` is 1 if cat is the top image. If you only report recall@1 on 100k images, a model that puts the right photo at rank 3 looks like a zero.

5. Where students get stuck

- Throwing away one tower and then being unable to query from that side.
- Reporting a single recall@1 without k or gallery size.
- Skipping a bias probe because "it's just a toy."

6. Video

Watch the **second half** of [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Start around **robustness to data shift (37:40)** and **broader impact (44:20)**. Pause on prompt sensitivity and on who is harmed when retrieval ranks people. Pair that with Lab 5's occupation probes (`nurse / ceo` vs `man / woman`), not with a vibe check.

7. Practice

1. You can retrieve with a caption but not with a photo. What did you probably throw away?
2. Why is recall@1 a poor single number for a 100k image gallery?
3. Write one probe prompt you would be embarrassed to ship, and what you would count as a bad retrieval.
4. Cosines of a query to four images: 0.11, 0.40, 0.39, 0.38. What is recall@1? recall@3? If the gold item is the 0.40 image, both are 1; if gold is 0.38, write both numbers.
5. Vectors $q = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $a = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$, $b = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$ (not yet normalized). Rank a, b by cosine to q . Then ℓ_2 -normalize a and b and rank again. Did the order change?